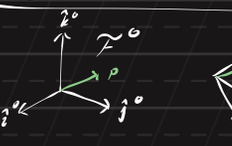


Chapter 2



$\hat{i}^2, \hat{j}^2, \hat{k}^2$ ($\cdot^i$ denotes frame)
Mutually orthogonal bases

There are different frames and we want to express a vector in one frame using the other

In $\mathcal{F}^0$ we have $\vec{p} = p_x^0 \hat{i}^0 + p_y^0 \hat{j}^0 + p_z^0 \hat{k}^0$
In $\mathcal{F}^1$ we have $\vec{p} = p_x^1 \hat{i}^1 + p_y^1 \hat{j}^1 + p_z^1 \hat{k}^1$

$$\Rightarrow p_x^0 \hat{i}^0 + p_y^0 \hat{j}^0 + p_z^0 \hat{k}^0 = p_x^1 \hat{i}^1 + p_y^1 \hat{j}^1 + p_z^1 \hat{k}^1$$

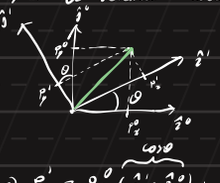
Doing a dot prod on each side with the bases of $\mathcal{F}^1$:

$$p_x^1 \underbrace{\hat{i}^1 \cdot \hat{i}^1}_{1} + p_y^1 \underbrace{\hat{j}^1 \cdot \hat{i}^1}_{0} + \ldots = (p_x^0 \hat{i}^0 + \ldots) \cdot \hat{i}^1$$

going from frame 0 to 1

$$\Rightarrow \vec{p}^1 = \begin{bmatrix} p_x^1 \\ p_y^1 \\ p_z^1 \end{bmatrix} = \begin{bmatrix} \hat{i}^1 \cdot \hat{i}^0 & \hat{i}^1 \cdot \hat{j}^0 & \hat{i}^1 \cdot \hat{k}^0 \\ \hat{j}^1 \cdot \hat{i}^0 & \hat{j}^1 \cdot \hat{j}^0 & \hat{j}^1 \cdot \hat{k}^0 \\ \hat{k}^1 \cdot \hat{i}^0 & \hat{k}^1 \cdot \hat{j}^0 & \hat{k}^1 \cdot \hat{k}^0 \end{bmatrix} \begin{bmatrix} p_x^0 \\ p_y^0 \\ p_z^0 \end{bmatrix} \Rightarrow \vec{p}^1 = R_0^1 \vec{p}^0$$

For a rotation about $\hat{k}^0 = \hat{k}^1$:

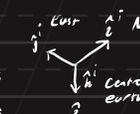


Basically projecting the original basis onto the new basis gives the factors

$$\Rightarrow p_x^1 = p_x^0 \underbrace{(\hat{i}^1 \cdot \hat{i}^0)}_{\cos\theta} + p_y^0 \underbrace{(\hat{i}^1 \cdot \hat{j}^0)}_{\sin\theta}$$

$$p_y^1 = p_x^0 \underbrace{(\hat{j}^1 \cdot \hat{i}^0)}_{-\sin\theta} + p_y^0 \underbrace{(\hat{j}^1 \cdot \hat{j}^0)}_{\cos\theta}$$

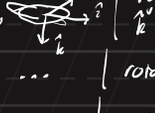
Frames:

$\mathcal{F}^i$: inertial, earth axes — 

NED

$\mathcal{F}^v$: Vehicle, coincides for pos. — 

aligned w NED

$\mathcal{F}^{v1}$: Vehicle frame 1, coincides for heading — 

rotation about $\hat{k}^v$ by heading $\Rightarrow \vec{p}^{v1} = R_v^{v1}(\psi)\vec{p}^v = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \vec{p}^v$

$\mathcal{F}^{v2}$: Vehicle frame 2, coincides for pitch ... — rotation about $\hat{j}^{v2} \Rightarrow p^{v2} = R_{v1}^{v2}(\theta)\vec{p}^{v1} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \vec{p}^{v1}$

$\mathcal{F}^b$: body, coincides for roll ... — rotation about $\hat{i}^{v2} \Rightarrow p^b = R_{v2}^b(\phi)\vec{p}^{v2} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \vec{p}^{v2}$

$$\Rightarrow \vec{p}^b = R_{v2}^b(\phi) R_{v1}^{v2}(\theta) R_v^{v1}(\psi) \vec{p}^v$$

$\mathcal{F}^s$: Stability, $\hat{i}$ is "aligned" with flow, for angle of attack — 

$\mathcal{F}^w$: wind, for sideslip

Velocity:

V_g – inertial velocity (ground vel)
V_a – Vel relative to air
V_w – wind velocity

$$\vec{V}_a = \vec{V}_g - \vec{V}_w$$

$$\Rightarrow V_g^b = \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$ inertial velocity in body frame

$$V_w^b = \begin{bmatrix} u_w \\ v_w \\ w_w \end{bmatrix} = R_v^b(\phi, \theta, \psi) \begin{pmatrix} w_n \\ w_e \\ w_d \end{pmatrix}$$ wind velocity in body frame

$$V_a^b = \begin{bmatrix} u_r \\ v_r \\ w_r \end{bmatrix} = \begin{bmatrix} u - u_w \\ v - v_w \\ w - w_w \end{bmatrix}$$ relative airspeed in body frame

From a similar analysis to above, using the definition of V_a and rotating to the body frame:

$$\begin{bmatrix} u_r \\ v_r \\ w_r \end{bmatrix} = V_a \begin{bmatrix} \cos\alpha\cos\beta \\ \sin\beta \\ \sin\alpha\cos\beta \end{bmatrix}$$ Very important for Physics

$$\Rightarrow V_A = \sqrt{u_r^2 + v_r^2 + w_r^2}$$

$$\alpha = \tan^{-1}\left(\frac{w_r}{u_r}\right)$$

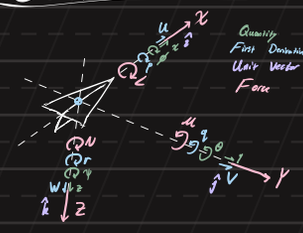
$$\beta = \sin^{-1}\left(\frac{v_r}{\sqrt{u_r^2 + v_r^2 + w_r^2}}\right)$$

Derivatives

$$\frac{d}{dt_i}\vec{p} = \frac{d}{dt_b}\vec{p} + \vec{\omega}_{b/i} \times \vec{p}$$ $\omega_{b/i}$: angular vel of body frame relative to inertial

1.) Set up simple rendering
2.) Implement rotation functions $\Rightarrow$ one function taking angles and outputting rotation matrix
3.) Test translation and rotation

Chapter 3



Kinematic relations

$$\frac{d}{dt}\begin{bmatrix} p_n \\ p_e \\ p_d \end{bmatrix} = R_b^v \begin{bmatrix} u \\ v \\ w \end{bmatrix} = (R_v^b)^T \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

each angle is in addition FOR so we diff in that FOR and then rotate

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} \dot\phi \\ 0 \\ 0 \end{bmatrix} + R_{v_1}^b(\phi)\begin{bmatrix} 0 \\ \dot\theta \\ 0 \end{bmatrix} + R_{v_2}^b(\phi) R_{v_1}^{v_2}(\theta)\begin{bmatrix} 0 \\ 0 \\ \dot\psi \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & -\sin\theta \\ 0 & \cos\phi & \sin\phi\cos\theta \\ 0 & -\sin\phi & \cos\theta\cos\theta \end{bmatrix}\begin{bmatrix} \dot\phi \\ \dot\theta \\ \dot\psi \end{bmatrix}$$

$$\begin{bmatrix} \dot p_n \\ \dot p_e \\ \dot p_d \end{bmatrix} = \begin{bmatrix} \cos\theta\cos\psi & \sin\phi\sin\theta\cos\psi - \cos\phi\sin\psi & \cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi \\ \cos\theta\sin\psi & \sin\phi\sin\theta\sin\psi + \cos\phi\cos\psi & \cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi \\ -\sin\theta & \sin\phi\cos\theta & \cos\phi\cos\theta \end{bmatrix}\begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

$$\begin{bmatrix} \dot\phi \\ \dot\theta \\ \dot\psi \end{bmatrix} = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi\sec\theta & \cos\phi\sec\theta \end{bmatrix}\begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

Dynamics

Translation

Using $V_g^b = [u\ v\ w]^T$, the velocity with respect to ground in the body frame

$$m\frac{d\vec{V}_g}{dt_i} = \underline{f} \quad \Big| \text{ we want in body frame} \Rightarrow = \frac{d\vec{V}_g}{dt_b} + \vec{\omega}_{b/i} \times \vec{V}_g$$

$\vec{\omega}_{b/i}$ from earlier is $[\dot\phi, \dot\theta, \dot\psi]^T$

Alternatively:

$$m\left(\frac{d\vec{V}_g^b}{dt_b} + \vec{\omega}_{b/i}^b \times \vec{V}_g^b\right) = \vec{f}^b \quad \text{which is in the body frame, } V_g^b = [u,v,w]^T,\ \omega_{b/i}^b = [p,q,r]$$

$$\frac{d\vec{V}_g}{dt_b} = \begin{bmatrix} \dot u \\ \dot v \\ \dot w \end{bmatrix} \Rightarrow \begin{bmatrix} \dot u \\ \dot v \\ \dot w \end{bmatrix} = \begin{bmatrix} rv - qw \\ pw - ru \\ qu - pv \end{bmatrix} + \frac{1}{m}\begin{bmatrix} f_x^b \\ f_y^b \\ f_z^b \end{bmatrix}$$

Rotational Motion

$$\frac{d\vec{L}}{dt_i} = \vec{\tau} = \frac{d\vec{L}}{dt_b} + \vec{\omega}_{b/i} \times \vec{L} \quad \Big| \text{ Assumes moments taken about COM}$$

($\vec{L}$ ← angular momentum)

Expressing in body frame:

$$\frac{d\vec{L}^b}{dt_b} + \vec{\omega}_{b/i}^b \times \vec{L}^b = \vec{\tau}^b \quad \Big| \quad I = \begin{bmatrix} I_x & I_{xy} & I_{xz} \\ & I_y & I_{yz} \\ & & I_z \end{bmatrix},\ \frac{dI}{dt} = 0$$

$$\vec{L}^b = \underline{I}^b \vec{\omega}_{b/i} \Rightarrow \frac{d\vec{L}^b}{dt_b} = \underline{I}\frac{d\vec{\omega}_{b/i}^b}{dt_b}$$

$$\Rightarrow \underline{I}\frac{d\vec{\omega}_{b/i}^b}{dt_b} + \vec{\omega}_{b/i}^b \times (\underline{I}\,\vec{\omega}_{b/i}^b) = \vec{\tau}^b \quad \Big| \quad \dot{\vec{\omega}}_{b/i}^b = \begin{bmatrix} \dot p \\ \dot q \\ \dot r \end{bmatrix} \text{ and } \dot\omega_{b/i}^b = \underline{I}^{-1}\left[-\omega_{b/i}^b \times (\underline{I}\,\vec{\omega}_{b/i}^b) + \vec{\tau}^b\right]$$

Using Symmetry:

$$I = \begin{bmatrix} I_x & 0 & -I_{xz} \\ 0 & I_y & 0 \\ -I_{xz} & 0 & I_z \end{bmatrix} \Rightarrow \underline{I}^{-1} = \begin{bmatrix} \frac{I_z}{\Gamma} & 0 & \frac{I_{xz}}{\Gamma} \\ 0 & \frac{1}{I_y} & 0 \\ \frac{I_{xz}}{\Gamma} & 0 & \frac{I_x}{\Gamma} \end{bmatrix} \quad \text{with } \Gamma = I_x I_z - I_{xz}^2$$

$$\Rightarrow \begin{bmatrix} \dot p \\ \dot q \\ \dot r \end{bmatrix} = \begin{bmatrix} \frac{I_z}{\Gamma} & 0 & \frac{I_{xz}}{\Gamma} \\ 0 & \frac{1}{I_y} & 0 \\ \frac{I_{xz}}{\Gamma} & 0 & \frac{I_x}{\Gamma} \end{bmatrix}\left(\begin{bmatrix} 0 & r & -q \\ -r & 0 & p \\ q & -p & 0 \end{bmatrix}\begin{bmatrix} I_x & 0 & -I_{xz} \\ 0 & I_y & 0 \\ -I_{xz} & 0 & I_z \end{bmatrix}\begin{bmatrix} p \\ q \\ r \end{bmatrix} + \begin{bmatrix} L \\ M \\ N \end{bmatrix}\right)$$

$$= \begin{bmatrix} \frac{I_z}{\Gamma} & 0 & \frac{I_{xz}}{\Gamma} \\ 0 & \frac{1}{I_y} & 0 \\ \frac{I_{xz}}{\Gamma} & 0 & \frac{I_x}{\Gamma} \end{bmatrix}\left(\begin{bmatrix} I_{xz}pq + (I_y - I_z)qr \\ I_{xz}(r^2 - p^2) + (I_z - I_x)pr \\ (I_x - I_y)pq - I_{xz}qr \end{bmatrix} + \begin{bmatrix} L \\ M \\ N \end{bmatrix}\right)$$

$$= \begin{bmatrix} \frac{I_{xz}(I_x - I_y + I_z)}{I_x I_z - I_{xz}^2}pq - \frac{I_z(I_z - I_y) + I_{xz}^2}{I_x I_z - I_{xz}^2}qr + \frac{I_z}{I_x I_z - I_{xz}^2}L + \frac{I_{xz}}{I_x I_z - I_{xz}^2}N \\ \frac{I_z - I_x}{I_y}pr - \frac{I_{xz}}{I_y}(p^2 - r^2) + \frac{M}{I_y} \\ \frac{(I_x - I_y)I_x + I_{xz}^2}{I_x I_z - I_{xz}^2}pq - \frac{I_{xz}(I_x - I_y + I_z)}{I_x I_z - I_{xz}^2}qr + \frac{I_{xz}}{I_x I_z - I_{xz}^2}L + \frac{I_x}{I_x I_z - I_{xz}^2}N \end{bmatrix}$$

$$= \begin{bmatrix} \frac{I_{xz}(I_x - I_y + I_z)}{I_x I_z - I_{xz}^2}pq - \frac{I_z(I_z - I_y) + I_{xz}^2}{I_x I_z - I_{xz}^2}qr \\ \frac{I_z - I_x}{I_y}pr - \frac{I_{xz}}{I_y}(p^2 - r^2) \\ \frac{(I_x - I_y)I_x + I_{xz}^2}{I_x I_z - I_{xz}^2}pq - \frac{I_{xz}(I_x - I_y + I_z)}{I_x I_z - I_{xz}^2}qr \end{bmatrix} + \begin{bmatrix} \frac{I_z}{I_x I_z - I_{xz}^2}L + \frac{I_{xz}}{I_x I_z - I_{xz}^2}N \\ \frac{M}{I_y} \\ \frac{I_{xz}}{I_x I_z - I_{xz}^2}L + \frac{I_x}{I_x I_z - I_{xz}^2}N \end{bmatrix}$$

1.) We can use $\begin{bmatrix} \dot p_n \\ \dot p_e \\ \dot p_d \end{bmatrix} = \dots$, $\begin{bmatrix} \dot u \\ \dot v \\ \dot w \end{bmatrix} = \dots$, $\begin{bmatrix} \dot\phi \\ \dot\theta \\ \dot\psi \end{bmatrix} = \dots$ and $\begin{bmatrix} \dot p \\ \dot q \\ \dot r \end{bmatrix} = \dots$ to represent our aircraft

2.) Implement the different formulations using both hand calcs and the full vector equations